

Assignment1Q2

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Question 2

(i) Residual Mean Square

The residual mean square for the fitted model is

$$MS_{Res} = 783.2559.$$

Relevant R output:

```
> summary(model)$sigma^2  
[1] 783.2559
```

(ii) Multiple Linear Regression Model

The fitted multiple linear regression model is

$$\widehat{gamble} = 69.222 - 33.752(sex) - 0.404(status) - 2.704(verbal).$$

The coefficient estimates were:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	69.22165	17.56194	3.94157	0.000293
sex	-33.75202	9.68393	-3.48536	0.001144
status	-0.40392	0.32665	-1.23654	0.222971
verbal	-2.70367	2.67838	-1.00944	0.318410

The coefficient for sex is -33.752. Sex is coded as 0 for males and 1 for females. Holding status and verbal score constant, females are predicted to spend approximately 33.75 units less on gambling than males.

The coefficient for status is -0.404. Holding the other predictors constant, a 1 unit increase in status resulted in an estimated decrease of 0.404 units in gambling expenditure.

The coefficient for verbal score is -2.704. Holding the other predictors constant, a 1 unit increase in verbal score resulted in an estimated decrease of 2.704 units in gambling expenditure.

The model has

$$R^2 = 0.2629.$$

Therefore, approximately 26.3% of the variation in gambling expenditure is explained by sex, status, and verbal score.

The overall F-test was

$$F(3, 43) = 5.111, \quad p = 0.004106.$$

Since the p-value is less than 0.05, the overall regression model is statistically significant. This indicates that at least one of the predictors is significantly associated with gambling expenditure.

Sex is statistically significant with a p-value of 0.00114. Status and verbal score are not statistically significant at the 5% significance level.

(iii) Multicollinearity

The VIF values were:

sex	status	verbal
1.355231	1.867493	1.452163

VIF values below 5. Thus, no evidence of problematic multicollinearity among sex, status & verbal score.

(iv) Confidence Intervals

The 95% confidence interval for sex is

$$(-53.282, -14.223).$$

The 95% confidence interval for verbal score is

$$(-8.105, 2.698).$$

Relevant R output:

	2.5 %	97.5 %
sex	-53.281523	-14.222509
verbal	-8.105134	2.697798

(v) OLS vs. Robust Regression

The OLS model produced

$$RSS_{OLS} = 33680.01$$

and

$$RMSE_{OLS} = 26.77.$$

The robust regression model produced

$$RSS_{Robust} = 39792.25$$

and

$$RMSE_{Robust} = 29.10.$$

OLS model provides better fit on the observed data

	RSS	RMSE
OLS	33680.01	26.76931
Robust	39792.25	29.09714

(vi) Outliers

An OLS model was fitted using gambling expenditure as the response variable and sex and status as predictors.

Externally studentized residuals were calculated and observations with

$$|r_i^*| > 2$$

were considered potential outliers.

The R output identified observation 24:

```
> studentRes[abs(studentRes) > 2]
      24
5.34369
```

Observation 24 had a studentized residual of

$$5.34369,$$

which is greater than 2 in absolute value.

The observation was:

	sex	status	income	verbal	gamble
24	0	27	10	4	156

Therefore, observation 24 is identified as an outlier. Its gambling expenditure of 156 is unusually high compared with the value predicted by the OLS model based on sex and status.

No other observations had an absolute studentized residual greater than 2.

Appendix: R Code

```

# install packages

library(faraway)
library(car)
library(MASS)

data(teengamb)

# show data
head(teengamb)
str(teengamb)

# Q2 i & ii)
# define mlr model & fit to predict
model <- lm(gamble ~ sex + status + verbal, data = teengamb)

summary(model)

# anova table
anova(model)

# get residual mean square
summary(model)$sigma^2

# coefficient table
coef(summary(model))

# Q2 iii) get multicollinearity
vif(model)

# Q2 iv) get 95% confidence intervals for B1(sex) & B3(verbal score)
confint(model, c("sex", "verbal"), level = 0.95)

# Q2 v) OLS vs robust regression

robustModel <- rlm(gamble ~ sex + status + verbal, data = teengamb)
summary(robustModel)

# compare coefficients of models
# show
coef(model)
coef(robustModel)

# compare rss

olsRSS <- sum(resid(model)^2)
robustRSS <- sum(resid(robustModel)^2)

# show
olsRSS

```

```

robustRSS

# compare RMSE
olsRMSE <- sqrt(mean(resid(model)^2))
robustRMSE <- sqrt(mean(resid(robustModel)^2))

# show
olsRMSE
robustRMSE

# Q2 vi)

outlierModel <- lm(gamble ~ sex + status, data = teengamb)

summary(outlierModel)

# studentized residues
studentRes <- rstudent(outlierModel)
studentRes

# show observations where absol studentized residue > 2
studentRes[abs(studentRes) > 2]

# display actual observations
teengamb[abs(studentRes) > 2,]

```